1-Which one of the notation that shown on the table bellow, when found on a chemical bottle indicates that this chemical is Corrosive :

A	B	C	D	E
				

2-What is the volume of 30 grams of lead (Pb) : [Pb density = 11.29 g/ml] environment

A) 4.4 ml B) 2.7 ml C) 8.9 ml D) 10.6 ml E)7.1 ml

3-An example of a physical property of chemical substance is :

A) Heat of dissociation B) Boiling point C) Heat of formation D) oxidation state E) Acidity

4-A 3.25g sample of hydrated sodium carbonate,$Na_2Co_3.xH_2O$ was placed into a crucible and heated, cooled and reweighed, the mass of the dehydrated sample was 1.35g, the value of water of crystallization x in the sample:

(Mm H 2O = 18 - Mm Na2CO3 = 105.97 g/mol)

A) 9 B) 6 C) 5 D) 3 E) 10

5- Combustion of a 0.9835-g sample of a compound containing only carbon, hydrogen, and oxygen produced 1.900 g of CO_2 and 1.070 g of H_2O. What is the empirical formula of the compound: (At.wts[g/mol] : H=1,O=16,C=12)

A) $C_4H_{10}O$ B) $C_4H_{11}O_2$ C) $C_4H_{10}O_2$ D) $C_2H_5O_2$ E) C_2H_5O

6-At which of the following pressure the boiling point of water is the highest :

A) 1atm B) 0.9 atm C) 1.2 atm D) 1.8 atm E) 720 mmHg

7-For the experiment of molecular weight of a volatile liquid, use the data in the following table to calculate the molecular weight of the volatile liquid in (g/mol) :

A) 194.4 B) 97,2
C) 486 D) 388.7
E)291.5

Mass of condensed liquid	0.25g
The temperature of the water in the water bath after complete vaporization of the volatile liquid	86℃
Atmospheric pressure	720 mmHg
Volume of flask used to vaporize the unknown volatile liquid	80 ml

8-The solution which filled in the burette in the acid base titration (determination of acetic acid in vinegar) is :

A) NaOCl B) HCl C) CH_3CO_2H D) NaOH E) $Na_2S_2O_3$

9- 2 ml of vinegar diluted with 14 ml of H_2O and titrated with 25ml of 0.08M NaOH solution to reach the end point ,the molar concentration of the diluted vinegar solution equal :

A) 0.1 M B) 0.4 M C) 0.125 M D) 0.08 M E) 0.2M

10- In the experiment of bleach analysis the data in the table were collected use them to calculate %W/W of NaOCl in bleach :

A) 1.67 % B) 5.0 % C) 10.0%
D) 1.25 % E) 2.5%

Volume of bleach sample	2ml
Volume of 0.1M Na2S2O3 solution	29 ml
Density of bleach	1.08g/ml
Molar mass of NaOCl	74.5g/mol

11-The indicator used in the bleach analysis is :

A) Vinegar B) phenolphthalein C) NaOCL D) starch E) acetic acid

12- A cup calorimeter of 22.0g mass was used to measure the heat evolved when 300ml of 0.1 M of NaOH was added to 1000 mL of 0.0300 M HNO3 initially at 23.5 oC. The temperature of the rxn rose to 30.5 oC and calorimeter and the . Calculate ΔH (in kJ/mol NaNO3) for this reaction. Assume the specific heat of the final solution is 4.18 J/g.C; the density of solution is 1.00 g/mL; heat capacity of cup = 92.5 J/c

$$HNO_3(aq) + NaOH(s) \longrightarrow NaNO_3(aq) + H_2O(l)$$

A) -1531kJ/mol B) -1392 kJ/mol C) -1289.5 kJ/mol D) -1810 kJ/mol E) -1670 kJ/mol

13-What is the heat capacity of 10g of Aluminum (CP= S= 0.90 J/g°C):

A) 11.7J/°C B) 6.3 J/°C C) 9.0 J/°C D)8.1 J/°C E) 13.5 J/°C

14) ΔH for the reaction : $IF_{5\,(g)} \longrightarrow IF_{3\,(g)} + F_{2\,(g)}$ is _____ kJ/mol, given the data below.

$IF_{(g)} + F_{2\,(g)} \longrightarrow IF_{3\,(g)}$ Δ H = -390 kJ/mol

$IF_{(g)} + 2\,F_{2\,(g)} \longrightarrow IF_{5\,(g)}$ Δ H = -745 kJ/mol

A) -1135 B) +35 C) +1135 D) -355 E) +355

15) Which of the following can be determined using the method used in Thermodynamic experiment {cup calorimeter (constant pressure calorimeter)}:

(1) Enthalpy change of combustion (2) Enthalpy change of dissociation

(3) Enthalpy change of formation

A) (1) and (2) only B) (1) and (3) only C) (2) and (3) only D) (1), (2) and (3) E) (1) only

16) By using Hess's Law,

A) The enthalpies of products in a reaction can be determined.

B) The enthalpy changes of reactions that cannot be performed easily can be found.

C) The enthalpy of reactant can be determined

D) A and C E) B and C

17) The rate law of the reaction $2NO_g + O_{2g} \longrightarrow NO_{2g}$ is :

A) Rate =K $[NO]^X$ B) Rate =K$[O_2]^X$ C) Rate= K$[NO]^X [O_2]^y$

D) Rate =K $[NO_2]^x$ E) Rate = K[NO][NO_2]

18) Nitric oxide reacts with chlorine to form nitrosyl chloride, NOCl. Use the following

data to determine the rate order with respect to NO.

Run#	[NO] M	$[Cl_2]$ M	Initial Rate M/s
1	0.22	0.065	0.96
2	0.66	0.065	8.6
3	0.22	0.032	0.38

$NO + 1/2Cl_2 \rightarrow NOCl$

A) first order

B) seconed order

C) third order

D) fourth order

E) zero order

The table on the right shows the data collected for the reaction $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + H_2O$

Use them to answer questions 19-20

Run #	Molarity of $S_2O_3^{2-}$ M	Time S	K rate constant $M^{-1}S^{-1}$
1	0.1	100	$2 *10^{-2}$
2	0.5	20	$4 *10^{-2}$

19) The rate in M/S for run #1 :

A) $1.0* 10^{-3}$ B) $5* 10^{-4}$
C) $8.3* 10^{-4}$ D) $1.25* 10^{-2}$ E) $1.9* 10^{-2}$

20) if you told that run#1 performed at 5 °C and run#2 at 20 °C , the E_a (activation energy of reaction) in KJ/mol is :

A) 31.3 B) 36.4 C) 43.4 D)53.3 E) 69.0

Fill the blanks with the correct answer

1- The glassware used to determine the volume of a metal in density experiment is called Graduated cylinder

2- An example of an alum is $KAl(SO_4)_2 \cdot 12H_2O$. //$NaFe(SO_4)_2 \cdot 12H_2O$

3- Exothermic reactions have ΔH of Negative sign

4- The empirical formula of C_6H_6 is CH with molar mass of 13